Semi-Automatic Onboarding for Personalised Travel Recommendations: The Place & Purpose Case Study

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***Abstract*—A personalised travel planning engine can only be as good as its knowledge of the traveller, but the direct way to gain that knowledge, a long questionnaire, is exactly what drives users away before the product has proven anything. We present a semi-automatic onboarding design for Place & Purpose that replaces the form with photo signal, a swipe-based preference game, and a short round of adaptive follow-up questions — chosen per traveller to close as many of the remaining gaps in their profile as possible — and ask whether adding a further signal, the traveller's public social media presence, is worth the legal risk it introduces. Having built and deployed this onboarding as a working proof of concept on the Claude API, we evaluated the resulting profiles on two planning tasks using onboarding time, human review, and an LLM-as-judge protocol. The social-media-informed profile showed a modest, task-dependent advantage — stronger on the longer planning task than the shorter one — while privacy trust remained the weakest-rated part of the flow. We close with a legal reading of both scenarios under the GDPR and the Swiss nFADP, and weigh the practical trade-offs a Swiss company would face in choosing between them today.**

I. Introduction

Travellers cannot find the right trip, or the right three hours in an unfamiliar city, without doing the searching themselves. Today that search happens across a dozen browser tabs and platforms, or through generic top-ranked lists that read the same for every visitor, and the one obvious way to make it personal, asking the traveller directly, backfires: a questionnaire long enough to build a useful profile is also long enough to make them leave before the product has proven any value, and people are in any case poor at describing their own taste in the abstract [1].

Meanwhile, a traveller's phone gallery, calendar, public posts, and two minutes of conversation already hold a specific, behavioural record of what they actually seek out [1]. That is the premise this paper builds on: shrink the asking to a handful of well-chosen prompts, and read the rest from signal the traveller already carries.

The remainder of the paper follows that logic through to a working system. Section II sets out the product's value proposition and the segments it serves. Sections III and IV describe the onboarding flow, architecture, and prompting behind the deployed proof of concept. Section V reports how well it worked, in onboarding time, human review, and LLM-as-judge scoring. Section VI discusses deployment, Section VII the legal picture, and Section VIII the discussion and future work, before Section IX concludes.

II. Value Proposition and Market Fit

Place & Purpose is a Swiss startup building a single application for dreaming, planning, and booking travel, in place of the browser tabs and platforms travellers currently juggle [1]. Its planning engine is explainable by design: suggestions are shaped around who the traveller actually is, and shown with the reasoning behind them, rather than as an opaque ranked list, which is what actually closes the gap described in Section I.

That promise rests on onboarding that takes a few minutes and reads from behavioural signal rather than a form, on suggestions that carry a stated reason rather than a bare ranking, and on one profile that carries over from a seven-day trip to a free afternoon instead of starting over each time. The deployed proof of concept described in Section III is itself the clearest demonstration of this: a flow a reader can go through end to end, not a mock-up.

Two segments motivate the design. Hotels are the earlier adopters: they want to differentiate the guest experience beyond a generic printed list of nearby restaurants, and are willing to pilot a concierge feature that does that. Individual travellers, particularly those already comfortable letting an AI assistant handle small planning decisions, want suggestions that feel personal rather than indistinguishable from a search engine. Both segments share the same constraint: the first suggestion has to already feel personal, which is why so much of the product's value depends on how well onboarding works before any plan is generated.

III. System Architecture

A. Onboarding flow

The onboarding flow collects signal through four complementary modalities rather than a single form:

1) **Consent**, split by data source (photo library, Instagram handle, Google account) rather than a single blanket toggle, so that each permission maps to a specific, explainable use.
2) **Identity and account linking**, which also determines whether the user is onboarded under Scenario A (no public social presence) or Scenario B (public social profile available).
3) **Five photos from past trips**, used as a behavioral record of what the user already sought out, rather than what they say they like.
4) **A pairwise preference swipe** across three categories (places, food, activities), up to five rounds per category.
5) **A short adaptive questionnaire**, closing gaps the earlier steps could not establish, in particular the hard constraints (diet, budget, mobility, travel companions, time windows).

B. Prototype architecture

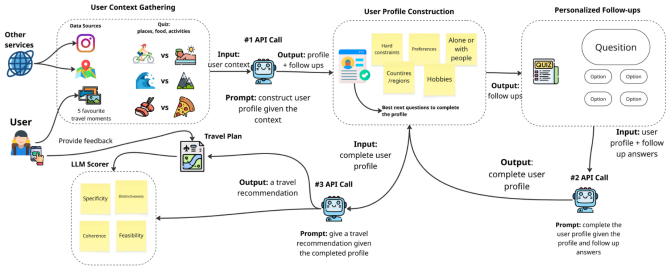


Fig. 1. Prototype architecture: three sequential LLM calls construct the traveller profile, generate targeted follow-up questions, and produce a travel recommendation, feeding into an LLM-based scorer.

The proof of concept was built and deployed as a Claude Artifact, with the Claude API as the backend for all three model calls (profile construction, gap-filling follow-ups, and recommendation generation).

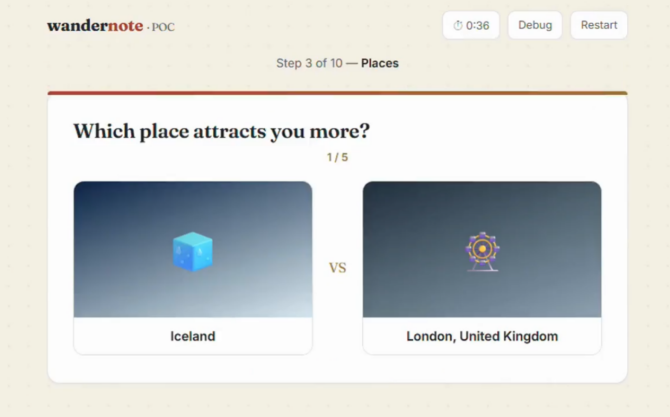


Fig. 2. Example of a demo page : would you rather phase.

IV. Model and Prompting

Between onboarding phases, the app makes three internal API calls to **Claude Sonnet 4.6**, kept identical across every experimental condition so that comparisons isolate the effect of the profile rather than the model [3].

TABLE I
DATA SOURCES BY ONBOARDING SCENARIO

Data source	Scenario A	Scenario B
On-device photo library	✓	✓
Pairwise preference swipe	✓	✓
Adaptive questionnaire	✓	✓
Public social media posts	–	✓

A. Three-stage prompting

1) *Profile construction and follow-up generation*: given the user's raw context (travel photos, swipe preferences, and optional social-media pictures), the model is prompted to analyse this data and populate an initial user profile across eleven fixed attributes. It then generates five personalised follow-up questions, targeted at the attributes the raw context left under-determined.
2) *Gap-filling*: given the partial profile and the user's answers to the five follow-up questions, a second call produces the completed profile.
3) *Recommendation*: given the completed profile, a third call produces a travel recommendation for the task the user selected (seven-day trip or three-hour Lausanne outing), under a fixed word limit and output structure so that profile-informed and profile-blind plans cannot be distinguished by length or verbosity alone [3].

V. Experimental Evaluation

Once built, the prototype needed to be tested to assess how well it performed and what adjustments it required. We used a two-phase evaluation method: a human-judge review of the onboarding experience itself, and an LLM-as-judge protocol to score the resulting trip recommendations at scale.

A. Human-judge review

Each of the three authors independently completed the onboarding flow and assessed whether the resulting recommendation matched their own profile, and whether any noticeable difference emerged between the no-social-media (Scenario A) and social-media (Scenario B) and conditions. Five criteria were scored on a 1–10 scale: personalisation/distinctiveness, trust/accuracy, usefulness, ease of use, and privacy comfort.

Table II reports a manual onboarding review across five UX criteria, scored by the group on its own onboarding sessions.

TABLE II
HUMAN-JUDGE ONBOARDING REVIEW

Metric	Score (avg of the 3 users)
Personalisation / distinctiveness	7 / 10
Trust / accuracy	8 / 10
Usefulness	8 / 10
Ease of use	10 / 10
Privacy comfort	6 / 10
Total	**78% (39/50)**

Across the three sessions, the authors did not observe a large perceptible difference between the plans produced under Scenario A and Scenario B: both were judged adequate and reasonably aligned with the stated profile. The main distinction noted was upstream of the recommendation itself, in the onboarding flow: under Scenario B, the follow-up questions generated after profile construction felt noticeably more personalised, drawing directly on details visible in the social-media context rather than on generic preference gaps. This suggests that, at this sample size, social-media data mainly sharpens the questions asked during onboarding.

B. LLM-as-judge scoring

An LLM judge can score many plans quickly, at a scale no small human panel could match within the timeframe of this project. Here, the goal was to compare the two trip recommendations, social-media-informed (B) against no-social-media (A), through two separate evaluation passes, deliberately kept apart to avoid conflating two different questions.

1) Primary pass: blind scoring: The judge sees only two anonymised plans, with no access to the underlying profile, and scores them on four metrics: specificity, coherence, feasibility, and distinctiveness. This avoids the circularity of asking whether a profile-aware model actually used the profile it was given [4]. The main goal of this pass is to evaluate the quality of the trip recommendation on its own terms.

2) Secondary pass: profile-aware grounded check: This pass answers a narrower question: not *which plan is better*, but *does this plan contradict the traveller it was built for?* The judge sees the profile alongside one plan, and lists contradictions against positively-supported elements. The main goal here is to test factual consistency between the plan's claims and what is actually known about the traveller, and to compare the two scenarios on that specific axis.

Known LLM-judge failure modes (verbosity bias, position bias, self-preference, and circularity) were mitigated by fixing output length across conditions, scoring each pair in both orders, and keeping the blind and grounded passes strictly separate [4].

Table III and IV reports the results of both passes, for the two type of itineraries.

TABLE III
LLM-SCORER RESULTS, SCENARIO A VS. SCENARIO B, 7 DAYS TRIP

Metric	**No social media (A)**	**Social media (B)**
Specificity	5	5
Coherence	4	5
Feasibility	4	4
Distinctiveness	4	5
Average blind score	4.25	**4.75**
Contradictions	3	2
Positively supported	6	7
Net grounded score	0	**3**

TABLE IV
LLM-SCORER RESULTS, SCENARIO A VS. SCENARIO B, 3 HOURS TRIP

Metric	**No social media (A)**	**Social media (B)**
Specificity	4	3
Coherence	3	3
Feasibility	4	2
Distinctiveness	3	3
Average blind score	**3.5**	2.75
Contradictions	2	1
Positively supported	5	5
Net grounded score	1	**3**

The picture differs by task length. For the seven-day trip (Table III), Scenario B scored slightly higher on the blind average (4.75 vs. 4.25) and showed fewer contradictions with more positively-supported claims, giving it the higher net grounded score. For the three-hour Lausanne outing (Table IV), this pattern partly reverses: Scenario A scored higher on the blind average (3.5 vs. 2.75), driven mainly by feasibility, yet Scenario B still produced fewer contradictions (1 vs. 2) despite an equal count of positively-supported elements. Taken together, the blind-quality advantage of social-media data does not hold consistently across task types, but its grounding advantage, fewer factual contradictions with what is actually known about the traveller, does. It is worth noting that this gap was measured under a conservative setup: Scenario B added only five manually uploaded social-media images on top of the same five gallery photos used in Scenario A, rather than a fully connected social profile.

C. Metrics Overview

Table V summarises the headline metrics measured across onboarding sessions, aggregating results from both planning tasks reported in tables III and IV.

TABLE V
ONBOARDING AND SATISFACTION METRICS

Average onboarding time	3 min 21 s (excl. model latency)
Overall UX satisfaction	78% (across 3 testers, 36–42/50)
Privacy trust	60% (lowest-rated, no variation)
Blind-condition distinguishability	4/5, naive judges could not reliably tell A from B
Contradiction reduction with social data	37% fewer, across 6 blind profile-fidelity comparisons
Short-task (3h) vs. long-task (7-day) blind score	3.6/5 vs. 4.5/5 average blind score

D. Interpretation

Taken together, the three evaluation phases point to a nuanced picture rather than a single verdict. The human-judge review (Section V-A) did not perceive a clear difference in the final plans themselves, though the follow-up questions generated under Scenario B felt noticeably more personalised. The LLM-scorer results (Section V-B) showed a modest, task-dependent advantage for Scenario B, stronger for the seven-day trip than the three-hour outing, and measured under

conditions likely to understate its true size (Section V-E). The three-hour task scored consistently lower than the seven-day task across both scenarios (3.1/5 vs. 4.5/5 average blind score), consistent with the case study's expectation that a short, local, time-boxed request leaves less room for a generic answer to pass unnoticed. Privacy trust remained the single weakest metric, at 60%, with no variation across testers, likely reflecting genuine discomfort with sharing social-media and photo data rather than random noise or a scoring artefact — a signal worth investigating further rather than dismissing.

E. Photo Signal Quality and Model Behaviour

The model's multimodal capabilities proved effective at extracting concrete signal from photos: it correctly identified a traditional Moroccan courtyard house (a *riad*) from architectural cues alone, and inferred an interest in winter sports from ski equipment visible in another image, alongside more general cues such as likely region, cultural or seasonal context, and clothing style. As authors, we deliberately uploaded a diverse, representative set of photos to maximise the information available to the model in *both* conditions, then added five further social-media images for Scenario B alone. This setup likely understates the real-world gap between scenarios in both directions: a typical user, less motivated to curate their camera roll, may upload repetitive or less informative gallery photos rather than searching for their most representative ones, while the five social-media images used here were themselves manually selected screenshots rather than a connected profile, drawing on a far smaller pool than the images, captions, and check-ins a real social media connection accumulates continuously over time. Under everyday conditions, the gap this evaluation measured could move in either direction: wider if an average user under-curates their gallery, or wider still in Scenario B's favour once it draws on a full, connected profile rather than five hand-picked images.

One recurring limitation in the model's reasoning was a tendency toward overly categorical inference from limited evidence. A traveller photographed alone was, in one instance, taken to prefer solo travel, though solitude in a single photo does not establish a general preference; similarly, not selecting "nightlife" in a pairwise comparison was read as an active dislike rather than simple indifference. Future prompt refinements should explicitly instruct the model to hedge such inferences proportionally to the evidence available, rather than treating a single data point as conclusive.

VI. MLOps and Production Deployment

The evaluated proof of concept was built and deployed as a Claude Artifact with the Claude API handling all three onboarding-side LLM calls end to end. This section separates that deployed state from the production architecture we would propose to take the prototype further, so as not to overstate what has actually been operationalised.

- **Data engineering**: version user-profile schemas and prompt templates with Git and DVC, so that a schema change never silently invalidates an existing profile; use an orchestration layer (e.g., n8n) to chain photo analysis, swipe merge, and gap-filling as small structured hand-offs between stages.
- **Deployment and monitoring**: containerise with Docker to keep prototype and production environments identical; track system metrics such as hallucination rate and confidence drift with Prometheus/Grafana.
- **Feedback loop**: a thumbs up/down on every suggestion as the simplest direct signal; session-level logging (e.g., PostHog) to trace a poor suggestion back to the exact profile that produced it.

VII. Ethics, Privacy, and Compliance

A. Scenario A: no public presence

Scenario A relies only on data the user actively grants: photo library access and gamified in-app questions. This is legal and low-risk under both the EU GDPR and the Swiss nFADP, since the lawful basis is direct, informed consent from the data subject for a clearly stated purpose.

B. Scenario B: public social profile

Scenario B adds public social media content as a further signal. Two access paths exist, with materially different legal standing:

- **Shadow scraping** of public pages: not legal in practice. It typically violates the platform's terms of service and does not, by itself, satisfy GDPR/nFADP requirements for processing personal data obtained from a third party.
- **Official platform API** access (e.g., Meta's API): a legal path, conditioned on scaling through an official developer request, a clear privacy policy, a verification step (e.g., a screencast of the user journey), and data minimisation, requesting only the permissions the feature actually needs.

C. Cross-border transfer to the LLM provider

Sending photos or profile data to an LLM API also constitutes an international data transfer (Switzerland/EU to the provider's servers). The relevant legal basis is the provider's Data Processing Agreement with Standard Contractual Clauses, accepted via the API's standard terms of use.

Data is not used to train the model by default, is retained for a bounded period for abuse monitoring, and users are informed in-app, before upload, that a third-party AI provider processes their data. This applies identically to both scenarios, since any uploaded photo passes through the same LLM pipeline regardless of source.

D. Risk of inferring sensitive characteristics

Beyond consent and lawful basis, both scenarios carry a subtler risk: the system may infer characteristics the traveller never disclosed. Dietary preferences can imply religion or a health condition; mobility constraints can imply disability; travel-companion patterns can imply sexual orientation or family status; avoided destinations can imply political views. Under GDPR Art. 9 and nFADP Art. 5(c), these count as

special-category data even when only inferred rather than stated, triggering a higher bar than ordinary consent. This risk is mitigated by design rather than by after-the-fact filtering: the profile schema excludes these categories entirely, and the profile-generation prompt treats exposure (a person appearing in or near something) as distinct from preference (liking it). Gallery and Instagram photos may also incidentally contain identifiable third parties who never consented to processing. The system does not include a facial-recognition component, and photos are used only for scene- and style-level inference about the account holder; no attempt is made to identify these individuals. This residual risk is disclosed to the uploader but does not constitute consent from anyone else depicted.

Gallery and Instagram photos may also incidentally contain identifiable third parties who never consented to processing. We do not perform facial recognition or attempt to identify these individuals; photos are used only for scene- and style-level inference about the account holder. This residual risk is disclosed to the uploader but does not constitute consent from anyone else depicted.

E. Verdict

Scenario A is deployable by a Swiss company today with standard consent practices. Scenario B is deployable only through the official API path, contingent on the platform granting the necessary permissions (e.g., Meta's developer review); the scraping path is not a legally sound option regardless of the perceived gain in personalisation.

VIII. Discussion and Future Work

Beyond the immediate limitations already noted (small sample size, an adaptive questionnaire whose branching should be verified rather than assumed, and a privacy-trust score that did not vary across testers), several extensions would meaningfully strengthen the product:

- **Maps integration**: real routes with actual distances, replacing the current text-only itinerary.
- **Speech recognition**: a voice note as an additional, lower-friction onboarding input.
- **Adaptive generative UI**: interface elements (e.g., quick intent filters) generated per traveller rather than fixed.
- **Itinerary builder**: auto-generate a full route from city, duration, and the existing profile.
- **Social layer**: sharing, browsing, and combining other travellers' routes.
- **Music as a behavioural signal**: connecting to the user's Spotify listening history as an additional, lower-friction signal than a full photo library, testing whether music preference correlates meaningfully with travel style. This remains speculative and would need its own validation before being treated as a reliable input.
- **Reinforcement learning from user feedback**: once thumbs up/down feedback (Section VI) accumulates at scale, use it as a reward signal to iteratively adjust the recommendation prompts (or fine-tune a scoring layer), rather than treating each rating as a one-off log entry. Over time, recommendations that were consistently rejected for a given profile type should become less likely, closing the loop between aggregate user feedback and the model's future outputs.

IX. Conclusion

We designed, built, and evaluated a semi-automatic onboarding flow for an AI travel planning engine, replacing a long-form questionnaire with photo signal, pairwise preference swiping, and a short adaptive questionnaire. The deployed proof of concept achieved a 3-minute-21-second onboarding time and 78% overall UX satisfaction, but privacy trust remained the weakest metric. Photos were only one of three complementary inputs: the pairwise preference swipe and the personalised follow-up questions each contribute their own piece of the picture, and both apply identically across Scenario A and Scenario B. The comparison below therefore isolates specifically the marginal value of adding social-media photos on top of an already-informative onboarding, not the onboarding flow as a whole.

Each scenario carries its own trade-offs, and our evaluation likely understates both in opposite directions. Scenario A is legally low-risk and deployable today with standard consent practices, but its quality depends entirely on what the user happens to upload: nothing in the flow prevents a less engaged user from supplying repetitive or uninformative photos, in which case the resulting profile degrades with no signal to the system that anything is wrong. Scenario B showed a higher blind score for the seven-day trip (4.75 vs. 4.25) and 37% fewer contradictions with the stated profile, but this was achieved with only five manually uploaded social-media screenshots, not a connected, fully analysed profile; a real social media connection, accumulating images, captions, and check-ins over time, would likely widen this gap further. This advantage was not uniform across tasks: for the shorter, three-hour outing, Scenario A scored higher on blind quality, though Scenario B still produced fewer contradictions with the stated profile. Against that upside, Scenario B is legally viable only through an official platform API (Section VII), which requires clearing a developer review process that not every team is positioned to pursue.

Thus, which scenario to prioritise depends on the available resources, risk appetite, and the ability to encourage users to upload informative photos (within Scenario A). Scenario A trades a lower barrier to deployment for sensitivity to user effort, while Scenario B trades a likely larger, though not yet fully measured, personalisation gain for materially higher legal and engineering overhead. The broader lesson holds regardless of that choice: reading a traveller's existing signal, rather than asking them to describe themselves, is what makes onboarding fast without making it shallow.

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